

FOR LOOPS

LEVEL 1: Loop Basics - Counting & Printing

Program 1: Print Numbers 1 to N Write a program that takes a number N as input and prints all numbers from 1 to N in a single line separated by spaces.

Program 2: Print Numbers N to 1 (Reverse) Write a program that takes a number N as input and prints all numbers from N to 1 in reverse order.

Program 3: Print First N Even Numbers Write a program that takes N as input and prints the first N even numbers (2, 4, 6, 8, ...).

Program 4: Print First N Odd Numbers Write a program that takes N as input and prints the first N odd numbers (1, 3, 5, 7, ...).

Program 5: Print Numbers Divisible by 5 (1 to N) Write a program that takes N as input and prints all numbers from 1 to N that are divisible by 5.

Program 6: Print Multiplication Table Write a program that takes a number N as input and prints its multiplication table from $N \times 1$ to $N \times 10$.

Program 7: Print Multiplication Table (Custom Range) Write a program that takes a number N and range (start to end) as input, and prints multiplication table of N for that range.

Program 8: Print Squares of First N Numbers Write a program that takes N as input and prints squares of first N natural numbers (1^2 , 2^2 , 3^2 , ..., N^2).

Program 9: Print Cubes of First N Numbers Write a program that takes N as input and prints cubes of first N natural numbers (1^3 , 2^3 , 3^3 , ..., N^3).

LEVEL 2: Summation & Calculation

Program 10: Sum of First N Natural Numbers Write a program that takes N as input and calculates the sum of first N natural numbers using a loop.

Program 11: Sum of First N Even Numbers Write a program that calculates the sum of first N even numbers using a loop.

Program 12: Sum of First N Odd Numbers Write a program that calculates the sum of first N odd numbers using a loop.

Program 13: Sum of Numbers Divisible by 3 (1 to N) Write a program that finds the sum of all numbers from 1 to N that are divisible by 3.

Program 14: Sum of Squares (First N Numbers) Write a program that calculates the sum of squares of first N natural numbers ($1^2 + 2^2 + 3^2 + \dots + N^2$).

Program 15: Sum of Cubes (First N Numbers) Write a program that calculates the sum of cubes of first N natural numbers ($1^3 + 2^3 + 3^3 + \dots + N^3$).

Program 16: Product of First N Numbers (Factorial) Write a program that takes N as input and calculates N! (factorial) using a loop.

Program 17: Power Calculator Write a program that takes base and exponent as input and calculates $\text{base}^{\text{exponent}}$ using a loop.

Program 18: Count Numbers Divisible by 7 (1 to N) Write a program that counts how many numbers from 1 to N are divisible by 7.

Program 19: Sum of All Divisors of N Write a program that takes a number N and calculates the sum of all its divisors (including 1 and N).

Program 20: Sum of Even and Odd Numbers (1 to N) Write a program that calculates sum of all even numbers and sum of all odd numbers from 1 to N separately.

Program 21: Find Largest Digit in a Number Write a program that takes a number and finds the largest digit in it using a loop.

Program 22: Find Smallest Digit in a Number Write a program that takes a number and finds the smallest digit in it using a loop.

LEVEL 3: Number Properties & Digit Manipulation

Program 23: Count Digits in a Number Write a program that takes a number as input and counts how many digits it has using a loop.

Program 24: Sum of Digits Write a program that takes a number as input and calculates the sum of its digits using a loop.

Program 25: Product of Digits Write a program that takes a number as input and calculates the product of its digits using a loop.

Program 26: Reverse a Number Write a program that takes a number as input and reverses it using a loop. Example: 1234 → 4321.

Program 27: Check if Palindrome Number Write a program that takes a number as input and checks if it is a palindrome using a loop.

Program 28: Armstrong Number Checker (Any Digits) Write a program that takes a number as input and checks if it is an Armstrong number using a loop.

Program 29: Find All Armstrong Numbers in Range Write a program that finds and prints all Armstrong numbers between 1 and N.

Program 30: Sum of First and Last Digit Write a program that takes a number and finds the sum of its first and last digits using a loop.

Program 31: Count Even and Odd Digits Write a program that takes a number and counts how many even digits and how many odd digits it has.

Program 32: Sum of Digits Until Single Digit (Digital Root) Write a program that keeps adding digits of a number until a single digit remains. Example: 9875 → 29 → 11 → 2.

Program 33: Print Digits Separately Write a program that takes a number and prints each digit on a new line.

Program 34: Check if Number Contains Specific Digit Write a program that takes a number and a digit as input, and checks if the number contains that digit.

Program 35: Count Occurrences of a Specific Digit Write a program that takes a number and a digit (0-9), and counts how many times that digit appears in the number.

Program 36: Remove All Occurrences of a Digit Write a program that takes a number and a digit, and creates a new number by removing all occurrences of that digit. Example: 12321, remove 2 → 131.

Program 37: Replace All Occurrences of a Digit Write a program that takes a number, an old digit, and a new digit, then replaces all occurrences of old digit with new digit. Example: 12321, replace 2 with 5 → 15351.

LEVEL 4: Prime Numbers & Factors

Program 38: Check if Prime Number Write a program that takes a number as input and checks if it is prime using a loop.

Program 39: Print All Prime Numbers (1 to N) Write a program that prints all prime numbers from 1 to N.

Program 40: Count Prime Numbers (1 to N) Write a program that counts how many prime numbers exist between 1 and N.

Program 41: Sum of All Prime Numbers (1 to N) Write a program that calculates the sum of all prime numbers from 1 to N.

Program 42: Print All Factors of a Number Write a program that takes a number as input and prints all its factors.

Program 43: Count Total Factors Write a program that takes a number as input and counts how many factors it has.

Program 44: Sum of All Factors Write a program that calculates the sum of all factors of a number (including 1 and the number itself).

Program 45: Sum of Proper Divisors Write a program that calculates the sum of all proper divisors of a number (excluding the number itself, including 1).

Program 46: Check Perfect Number Write a program that checks if a number is a perfect number (sum of proper divisors = number). Example: $6 = 1+2+3$.

Program 47: Find All Perfect Numbers (1 to N) Write a program that finds and prints all perfect numbers from 1 to N.

Program 48: Check Abundant Number Write a program that checks if a number is abundant (sum of proper divisors > number). Example: $12 \rightarrow 1+2+3+4+6=16 > 12$.

Program 49: Prime Factorization Write a program that takes a number and prints its prime factorization. Example: $60 \rightarrow 2 \ 2 \ 3 \ 5$ (print factors as they divide).

Program 50: Count Prime Factors Write a program that counts how many prime factors a number has (with repetition). Example: $60 \rightarrow 2, 2, 3, 5 \rightarrow \text{count} = 4$.